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723.ALLOGENEIC TRANSPLANTATION: LONG-TERM FOLLOW-UP, COMPLICATIONS, AND DISEASE RECURRENCE

Bacterial (Non-Clostridium Difficile) and Parasitic Stool Studies Have Limited Utility in Pediatric Hematopoietic Stem Cell Transplant Patients

Matthew Nagy, MDMPH^{1,2}, Chiara Wyckera, MD^{1,2}, Jeffrey Schemm, MD^{2,1}, Ryan Brewster, MD^{1,2}, Christine Duncan, MD^{3,4,5}

¹Boston Medical Center, Boston

²Boston Children's Hospital, Boston

³Department of Pediatric Oncology, Dana-Farber Cancer Institute, Boston, MA

⁴Department of Pediatrics, Harvard Medical School, Boston

⁵Division of Pediatric Hematology/Oncology, Boston Children's Hospital, Boston, MA

Background: Diarrhea is a common complication among pediatric hematopoietic stem cell transplantation (HCT) recipients. Although the majority of cases are secondary to graft versus host disease (GVHD), stool microbiological studies are often performed to evaluate an underlying infectious etiology. The aim of this study was to assess the frequency and utility of stool studies in children who have undergone HCT.

Methods: Demographics, clinical characteristics, and results for bacterial (*Shigella*, *Salmonella enterica*, *Yersinia enterocolitica*, *Enterohemorrhagic Escherichia coli* (SSYCE), and *Clostridium Difficile*), viral (Rotavirus, Cytomegalovirus, Norovirus, and Adenovirus), and parasitic (ova & parasite examination, of all patients who underwent HCT at a large, academic, freestanding children's hospital and affiliate cancer center and were admitted to the hospital for more than one day between Jan 2006 and Dec 2023 were obtained. Statistical analysis conducted included t-tests, chi-square and linear regression.

Findings: Overall, 1381 HCT recipients (9.2 ± 6.6 years) were included. Altogether, 6509 stool studies were obtained among 741 (54%) patients. Those with stool studies sent were more likely to have undergone an allogeneic transplant (63% vs 72%, $p < 0.001$). Of those with allogeneic transplant, those with stool studies sent had higher incidence of GVHD, both acute (27% vs 17%, $p = 0.001$) and chronic (23% vs. 11%, $p < 0.001$). SSYCE studies were sent on 363 (26%, 2252 studies) patients with one (0.04%) positive result for *Yersinia enterocolitica*. *Clostridium Difficile* was sent on 706 (51%, 2055 studies) patients, with 156 positive results (7.6%). Stool ova and parasite was sent on 143 (10%, 242 studies) patients, with two positive results of *Giardia lamblia* (0.8%), however, in both instances the studies prior to HCT showed positivity for the same parasite. Viral studies were sent on 638 (46%, 1960 studies) patients, with 107 positive studies (5.5%) Over the years, there was a significant increase in studies sent for yersinia ($p = 0.013$) and adenovirus ($p = 0.007$), and decrease for rotavirus ($p = 0.011$) and clostridium difficile ($p < 0.001$).

Interpretation: We identified a negligible rate of positivity for SSYCE and parasitic stool studies, despite the use of these tests in more than a quarter of all HCT patients admitted in the prior 8 years. Given the overall low diagnostic value, we argue that routine use of these studies in post-HCT pediatrics patients should be avoided, barring specific clinical circumstances, which would also serve to avoid unnecessary testing costs.

Disclosures No relevant conflicts of interest to declare.

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